

```
1 #include <stdio.h>
2
3 int factorial(int n) {
4     if (n == 0 || n == 1)
5         return 1;
6
7     return n * factorial(n - 1);
8 }
9
10 int main() {
11     int n;
12
13     printf("Enter a number: ");
14     scanf("%d", &n);
15
16     printf("Factorial = %d\n", factorial(n));
17
18     return 0;
19 }
```

```
1 #include <stdio.h>
2
3 int fibonacci(int n) {
4     if (n == 0)
5         return 0;
6     if (n == 1)
7         return 1;
8
9     return fibonacci(n - 1) + fibonacci(n - 2);
10 }
11
12 int main() {
13     int n;
14
15     printf("Enter n: ");
16     scanf("%d", &n);
17
18     printf("Fibonacci number = %d\n", fibonacci(n));
19
20     return 0;
21 }
```

```
1 #include <stdio.h>
2
3 int sumNatural(int n) {
4     if (n == 0)
5         return 0;
6
7     return n + sumNatural(n - 1);
8 }
9
10 int main() {
11     int n;
12
13     printf("Enter n: ");
14     scanf("%d", &n);
15
16     printf("Sum = %d\n", sumNatural(n));
17
18     return 0;
19 }
```

```
1 #include <stdio.h>
2
3 int gcd(int a, int b) {
4     if (b == 0)
5         return a;
6
7     return gcd(b, a % b);
8 }
9
10 int main() {
11     int a, b;
12
13     printf("Enter two numbers: ");
14     scanf("%d %d", &a, &b);
15
16     printf("GCD = %d\n", gcd(a, b));
17
18     return 0;
19 }
```

```
1 #include <stdio.h>
2
3 int power(int base, int exponent) {
4     if (exponent == 0)
5         return 1;
6
7     return base * power(base, exponent - 1);
8 }
9
10 int main() {
11     int base, exponent;
12
13     printf("Enter base and exponent: ");
14     scanf("%d %d", &base, &exponent);
15
16     printf("Power = %d\n", power(base, exponent));
17
18     return 0;
19 }
```

```
1 #include <stdio.h>
2
3 void reverse(char str[], int index) {
4     if (str[index] == '\0')
5         return;
6
7     reverse(str, index + 1);
8     printf("%c", str[index]);
9 }
10
11 int main() {
12     char str[100];
13
14     printf("Enter a string: ");
15     scanf("%s", str);
16
17     printf("Reversed string: ");
18     reverse(str, 0);
19
20     return 0;
21 }
```

```
1 #include <stdio.h>
2
3 int isPalindrome(char str[], int start, int end) {
4     if (start >= end)
5         return 1;
6
7     if (str[start] != str[end])
8         return 0;
9
10    return isPalindrome(str, start + 1, end - 1);
11 }
12
13 int main() {
14     char str[100];
15     int length = 0;
16
17     printf("Enter a string: ");
18     scanf("%s", str);
19
20     while (str[length] != '\0')
21         length++;
```

```
1 #include <stdio.h>
2
3 void swap(char *a, char *b) {
4     char temp = *a;
5     *a = *b;
6     *b = temp;
7 }
8
9 void permutations(char str[], int start, int end) {
10     int i;
11
12     if (start == end) {
13         printf("%s\n", str);
14         return;
15     }
16
17     for (i = start; i <= end; i++) {
18         swap(&str[start], &str[i]);
19         permutations(str, start + 1, end);
20         swap(&str[start], &str[i]);
21     }
```



```
1 #include <stdio.h>
2
3 void combinations(int arr[], int n, int index, int start) {
4     if (index == n) {
5         for (int i = 0; i < n; i++)
6             printf("%d ", arr[i]);
7         printf("\n");
8         return;
9     }
10
11     for (int i = start; i <= n; i++) {
12         arr[index] = i;
13         combinations(arr, n, index + 1, i + 1);
14     }
15 }
16
17 int main() {
18     int n;
19     int arr[100];
20
21     printf("Enter number of elements: ");
```

```
1 #include <stdio.h>
2
3 void towerOfHanoi(int n, char source, char auxiliary, char destination) {
4     if (n == 1) {
5         printf("Move disk 1 from %c to %c\n", source, destination);
6         return;
7     }
8
9     towerOfHanoi(n - 1, source, destination, auxiliary);
10
11     printf("Move disk %d from %c to %c\n", n, source, destination);
12
13     towerOfHanoi(n - 1, auxiliary, source, destination);
14 }
15
16 int main() {
17     int n;
18
19     printf("Enter number of disks: ");
20     scanf("%d", &n);
21
```

```
1 #include <stdio.h>
2
3 int sumDigits(int n) {
4     if (n == 0)
5         return 0;
6
7     return (n % 10) + sumDigits(n / 10);
8 }
9
10 int main() {
11     int n;
12
13     printf("Enter a number: ");
14     scanf("%d", &n);
15
16     printf("Sum of digits = %d\n", sumDigits(n));
17
18     return 0;
19 }
```

```
1 #include <stdio.h>
2
3 void geometricSeries(int term, int n, int a, int r) {
4     if (term > n)
5         return;
6
7     printf("%d ", a);
8
9     geometricSeries(term + 1, n, a * r, r);
10 }
11
12 int main() {
13     int n, a, r;
14
15     printf("Enter number of terms: ");
16     scanf("%d", &n);
17
18     printf("Enter first term: ");
19     scanf("%d", &a);
20
21     printf("Enter common ratio: ");
```

```
1 #include <stdio.h>
2
3 int arithmeticTerm(int a, int d, int n) {
4     if (n == 1)
5         return a;
6
7     return d + arithmeticTerm(a, d, n - 1);
8 }
9
10 int main() {
11     int a, d, n;
12
13     printf("Enter first term: ");
14     scanf("%d", &a);
15
16     printf("Enter common difference: ");
17     scanf("%d", &d);
18
19     printf("Enter n: ");
20     scanf("%d", &n);
21
```

```
1 #include <stdio.h>
2
3 int sumEven(int n) {
4     if (n == 0)
5         return 0;
6
7     return 2 * n + sumEven(n - 1);
8 }
9
10 int main() {
11     int n;
12
13     printf("Enter n: ");
14     scanf("%d", &n);
15
16     printf("Sum of first %d even numbers = %d\n", n, sumEven(n));
17
18     return 0;
19 }
```

```
1 #include <stdio.h>
2
3 void binary(int n)
4 {
5     if (n > 1)
6         binary(n / 2);
7
8     printf("%d", n % 2);
9 }
10
11 int main()
12 {
13     int n;
14
15     scanf("%d", &n);
16
17     binary(n);
18
19     return 0;
20 }
```

```
1 #include <stdio.h>
2
3 int length(char str[])
4 {
5     if (str[0] == '\0')
6         return 0;
7
8     return 1 + length(str + 1);
9 }
10
11 int main()
12 {
13     char str[100];
14
15     scanf("%99s", str);
16
17     printf("%d", length(str));
18
19     return 0;
20 }
```



```
1 #include <stdio.h>
2
3 int count(char str[], char ch)
4 {
5     if (str[0] == '\0')
6         return 0;
7
8     return (str[0] == ch) + count(str + 1, ch);
9 }
10
11 int main()
12 {
13     char str[100], ch;
14
15     scanf("%99s", str);
16     scanf(" %c", &ch);
17
18     printf("%d", count(str, ch));
19
20     return 0;
21 }
```

```
1 #include <stdio.h>
2
3 int sumArray(int a[], int n)
4 {
5     if (n == 0)
6         return 0;
7
8     return a[n - 1] + sumArray(a, n - 1);
9 }
10
11 int main()
12 {
13     int a[100], n, i;
14
15     scanf("%d", &n);
16
17     for (i = 0; i < n; i++)
18         scanf("%d", &a[i]);
19
20     printf("%d", sumArray(a, n));
21
```

```
1 #include <stdio.h>
2
3 int check(int n, int i)
4 {
5     if (i * i == n)
6         return 1;
7     if (i * i > n)
8         return 0;
9
10    return check(n, i + 1);
11 }
12
13 int main()
14 {
15     int n;
16
17     scanf("%d", &n);
18
19     if (check(n, 1))
20         printf("Perfect Square");
21     else
```

```
1 #include <stdio.h>
2
3 int reverse(int n, int rev)
4 {
5     if (n == 0)
6         return rev;
7
8     return reverse(n / 10, rev * 10 + n % 10);
9 }
10
11 int main()
12 {
13     int n;
14
15     scanf("%d", &n);
16
17     printf("%d", reverse(n, 0));
18
19     return 0;
20 }
```